Market Analysis — UART/mmWave DIY Kit (HLK-LD2410C radar + BH1750)

Analysis of the `uart_option` build sold as a **DIY kit** (no certification required initially). Prices in MXN, mid-range MercadoLibre México / Amazon MX (Sept 2026, ~1 USD = 18 MXN).

---

1. Product — the DIY kit

A **"build-it-yourself" radar stair-light kit**: two HLK-LD2410C mmWave radars (top/bottom), a BH1750 ambient-light sensor, and a main controller board, wired together with **standard Cat6 patch cables** via **RJ45 jacks** (no soldering, no I²C pull-up tuning). The customer mounts it, plugs it in, and gets a staircase that lights itself when someone steps on it — and stays on while they stand still.

Sold as a kit, not a finished appliance → in Mexico this generally sidesteps NOM-001-SCFI (finished-product electrical safety) and IFETEL (the ESP32 module is already homologated by its manufacturer). This removes the ~$60K certification cliff from the earlier analyses and is the whole point of the kit strategy.

Two SKUs

SKU	Contents	Retail
Core kit	main board + 2 radars + BH1750 + RJ45 adapters + Cat6 cables + instructions	**$1,499**
Complete kit	core kit + 5 V USB LED strip + USB charger + enclosures + hardware	**$1,899**

---

2. Market analysis

2a. Target market

Question	Answer
Who buys it?	Makers/hobbyists, Home Assistant / ESPHome users, semi-pro installers, tech-savvy homeowners
Problem solved	Safety + convenience on stairs, without paying for a "smart home" hub, and without a custom build
Why a kit?	Lower price than finished products, the satisfaction of DIY, and **no certification** = available now
Addressable (MX)	~600K automation-interested households + the maker community + LATAM ESPHome/HA users
Obtainable Year 1	150–400 kits (realistic)

2b. Competitors (kit/DIY framing)

Competitor	Price (MXN)	What it is	Weakness vs. us
Battery PIR stair lights (ML/Amazon)	$150–250	Stick-on, battery, motion-only	No still-person detection, recharging, no API
Smart-home hub kits (Tuya/Aqara)	$800–2,000	Hub + motion sensor + bulbs	Needs hub, single-sensor, cloud dependency
Finished radar presence (Aqara FP2, etc.)	$1,200–2,000	mmWave presence, off-the-shelf	Single sensor, closed ecosystem, not stair-specific
DIY-from-scratch (ESPHome/Tindie tutorials)	$800–1,300 (parts)	Build it yourself from a blog post	No curated kit, no instructions, fiddly I²C
Finished stair-light products	$1,699–2,999	Our own other variants (finished)	Requires certification to sell legally as a product
"Do nothing"	$0	manual switch	The eternal competitor

Key differentiator: we are the only **purpose-built staircase radar kit** with dual still-person detection + a zero-solder Cat6/RJ45 build. The "kit" framing also *unlocks the market now* (no certification lead time).

2c. SWOT

Strengths	Weaknesses
Dual mmWave detects still people (better than PIR/ToF)	Kit = customer does the assembly (support load)
Cat6/RJ45 — no soldering, foolproof wiring	Smaller "maker" market vs mass consumer
No certification needed to sell (kit)	Lower perceived polish vs finished product
Real lux (BH1750) + web dashboard + API	Requires a 5 V USB feed near the stairs

Opportunities	Threats
Growing Home Assistant / ESPHome community in LATAM	Cheap mmWave modules + free tutorials
"No-cert" kit → first-to-market speed	Regulatory interpretation of "kit" could shift
Upsell: complete kit, install service, pro SKU	Copycats on Tindie/AliExpress
Partnerships with maker stores (Tindie, local)	USD/MXN volatility on radar modules

---

3. Bill of Materials (with Cat6/RJ45 PCB connectors)

The Cat6 connection is handled by **RJ45 PCB-mount jacks** on the main board and a **radar adapter PCB** (RJ45 + 5-pin radar header) at each radar — so the customer just plugs in standard Ethernet patch cables.

3a. Core kit (small batch, ~100 units)

#	Component	Qty	Unit	Total
1	ESP32-S3 SuperMini	1	$200	$200
2	HLK-LD2410C mmWave radar	2	$180	$360
3	BH1750 ambient light sensor	1	$75	$75
4	IRLZ44N MOSFET (TO-220)	1	$30	$30
5	**RJ45 PCB-mount jack (Cat6)**	2	$15	$30
6	**Radar adapter PCB (RJ45 + 5-pin header)**	2	$30	$60
7	**Cat6 patch cable 3 m**	2	$30	$60
8	Screw terminals + passives (resistors)	1 set	$30	$30
9	Main enclosure	1	$80	$80
10	Radar enclosures	2	$40	$80
11	Instructions + packaging	1	$60	$60
12	Mounting hardware	1	$30	$30
	Core kit subtotal			**$1,095**

3b. Complete kit (adds)

#	Component	Qty	Unit	Total
13	5 V LED strip (USB-A, white)	1	$200	$200
14	5 V / 2 A USB charger	1	$120	$120
	Complete kit total			**$1,415**

3c. Mass production (custom PCB + bulk radar)

Scenario	Core kit	Complete kit
Small batch (100)	$1,095	$1,415
Mass production (1,000)	**~$460**	**~$560**

At scale the custom PCB integrates the ESP32-S3 module, BH1750, MOSFET and the two RJ45 jacks on one board (~$140 assembled — no regulator needed on a 5 V system); radar modules drop to ~$80 each in bulk.

4. Unit cost & unit economics

4a. Unit cost (COGS)

	Core kit	Complete kit
Small batch	$1,095	$1,415
Mass production	$460	$560

4b. Unit economics (direct sale)

```
Core kit Complete kit
Retail price: $1,499 $1,899
- COGS (mass prod): $460 $560
- Payment processing (3%): $45 $57
- Shipping: $120 $130
- Support: $20 $20
- Marketing (CAC): $50 $60
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
Net profit / unit: $804 $1,072
Gross margin: 53.6% 56.5%
```

Small-batch margins are thin (~11%) — the business case, as always, is the mass-production COGS.

---

5. Sell analysis — ROI & break-even

5a. Fixed costs (Year 1, DIY kit — NO certification)

Item	Cost (MXN)	Recurring?
PCB design + prototypes	$6,000	Once
Enclosure design + 3D prints	$4,000	Once
Instruction manual (design + print)	$6,000	Once
Website (Shopify)	$6,000	Annual
Domain + email	$1,200	Annual
Packaging design	$5,000	Once
Photography + video	$3,000	Once
Marketing (Meta/Google + community)	$36,000	$3,000/mo
Legal (trademark only)	$4,000	Once
Accounting / SAT	$6,000	Annual
Total fixed (Year 1)	**~$77,200**	

> ■ vs. **~$157,200** for the certified finished product — the kit strategy cuts fixed costs **~$80K** by skipping NOM/IFETEL.

5b. Break-even (mass-production COGS)

```
Fixed costs: $77,200
Gross profit (core kit): $804 → break-even ≈ 96 units
Gross profit (complete): $1,072 → break-even ≈ 72 units
```

Scenario	Units/mo	Break-even	Year 1 profit/loss
Optimistic (50/mo)	600/yr	~2 months	+$405K
Realistic (20/mo)	240/yr	~5 months	+$115K
Pessimistic (5/mo)	60/yr	~19 months	–$29K

---

6. Sell plan (DIY kit, no-cert strategy)

6a. Positioning

> ■ **"Build your own radar stair light — no soldering, no certification, no monthly fees."** A curated kit: plug Cat6 into RJ45, flash, mount, done.

The "kit" framing is the strategy — it ships *now* (no cert lead time), targets the growing maker/Home-Assistant audience, and sidesteps the regulatory cliff.

6b. Channels

Channel	Role	Notes
Tindie	Primary (maker marketplace)	The natural home for a no-cert kit
Direct / Shopify	Brand + margin	3–5% fees, own the data
MercadoLibre	LATAM reach	list as "kit de partes / educativo"
Amazon.com.mx	Trust	list as "kit/componentes"
Facebook/Reddit HA & ESPHome groups	Community + validation	zero-cost demand gen

6c. Go-to-market

1. Build 5 kits, place with Home Assistant / ESPHome community members.
2. Pre-sell on Tindie + a landing page ("radar stair kit") with $100 in ads.
3. If 10+ pre-orders → design the custom PCB (RJ45 + radar adapters).
4. 50-kit batch at $1,499 (core) / $1,899 (complete).
5. Scale; later offer a **certified "pro" finished version** as the upsell once the kit proves demand.

---

7. Verdict

	I²C finished (baseline)	**UART/mmWave DIY kit (this branch)**
BOM (mass prod)	$790	**$460 (core) / $560 (complete)**
Fixed costs	$157,200 (certified)	**$77,200 (no cert)**
Break-even	~322 units	**~72–96 units**
Retail	$1,699	$1,499 / $1,899
Still-person detection	narrow beam	**wider cone + micro-motion**
Wiring	I²C (address + pull-ups)	**Cat6/RJ45, no solder**

The DIY kit is the most capital-efficient way to market: it drops the certification cost, breaks even at ~a quarter of the certified product's volume, and reaches the maker/Home-Assistant audience *now* — while keeping the radar "still-person detection" differentiator. The trade-off is a smaller market and more support per sale; the mitigation is a certified "pro" SKU as the eventual upsell.